Algebra

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[0.1] Any errors introduced are my own.

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0.1 Introduction

These notes were taken during the course math 250A in fall 2017 at UC Berkeley. The professor was Richard Borcherds, to whom I defer credit for all of the mathematical content in this document. I myself claim ownership of all the errors in this document, of which there are likely many.

This course is meant to serve as an early graduate algebra course. Roughly speaking, the first part concerns itself over groups, the second moves to rings, modules, and polynomials, and the final portion focuses on field extensions and Galois theory. Though I made an effort to keep these notes logically self-contained, there will likely be some portions that assume some previous knowledge of algebra.

If you have any questions, suggestions, or comments concerning these notes, please do not hesitate to contact me at jacksontvandyke@gmail.com.

Chapter 1

Groups

1.1 Definitions

Definition 1.1 (Concrete group)**.** A set G is a group iff it is the set of symmetries of something.

Example 1.1. We can consider the symmetries of a rectangle to be: doing nothing, reflecting (horizontal and vertical), and rotating. We will be returning to this geometric interpretation later.

Definition 1.2 (Abstract group)**.** A set G with a binary relation (if $a, b \in G$, written $a \times b, a + b, a \cdot b, a \circ b$ or just ab) is a group iff this relation is associative, each element has an inverse, and there is an identity. In other words, for all $a, b, c \in G$,

1. There exists $e \in G$ such that $ea = ae = a$
2. There exists $a^{-1} \in G$ such that $aa^{-1} = a^{-1}a = e$
3. $a(bc) = (ab)c$

Remark 1.1. It is clear that if we take the composition of symmetries as our binary relation, then the concrete notion of a group can be translated to the abstract notion. It is a subtle and important point that the converse is true. First we consider some basic definitions.

Definition 1.3 (Subgroup)**.** Let G be a group. A subset $H \subseteq G$ is a *subgroup* of G iff it contains the identity of G, is closed under the law of composition, and contains all inverses.

Definition 1.4 (Homomorphism)**.** Let G, H be two groups. Then a function $f : G \to H$ is a homomorphism iff for all $a, b \in G$,

$$f(ab) = f(a) f(b) \tag{1.1}$$

Proposition 1.1. *Homomorphisms preserve inverses and the identity.*

Proof. Let G, H be groups, and take $f : G \to H$ to be a homomorphism. Then for all $g \in G$ we have that

$$f(g) = f(ge) = f(g) f(e) = f(eg) = f(e) f(g) \tag{1.2}$$

then by the definition, and uniqueness of the identity element, $f(e)$ is the identity in H. Then if we have $a \in G$ with inverse a^{-1} we can write

$$f(e) = f\left(aa^{-1}\right) = f(a) f\left(a^{-1}\right) = f\left(a^{-1}a\right) = f\left(a^{-1}\right) f(a) \tag{1.3}$$

so since f preserves the identity, we have that it also preserves inverses. □

Definition 1.5 (Isomorphism)**.** A homomorphism is an *isomorphism* iff it is a bijection.

Definition 1.6 (Endomorphism)**.** A homomorphism is an *endomorphism* iff it has the same domain as its codomain.

Definition 1.7 (Automorphism)**.** A homomorphism is an *automorphism* iff it is an isomorphism and an endomorphism.

Example 1.2. Let $G = \langle \mathbb{R}, + \rangle$ and $H = \langle \mathbb{R}^{\times}, \times \rangle$. Then if we take $f : G \to H$ to be the exponential map, we get an isomorphism.

Definition 1.8. Let G be a group and S a set. Then a map $\cdot : G \times S \to S$ is a *left action* of G on S iff for all $s \in S$ and all $g, h \in G$ we have:

1. $g \cdot (h \cdot s) = (gh) \cdot s$
2. $e \cdot s = s$

where e is the identity for G. A *right action* is a map $\cdot : S \times G \to S$ which satisfies the analogous properties.

Definition 1.9. Let G be a group. A set S is a G-set iff we have an action of G on S which is a homomorphism

$$\pi : G \to \operatorname{Perm}(S) \tag{1.4}$$

Example 1.3. Let G be a group, and S a G-set. Then the image of G in $\operatorname{Perm}(S)$ is a subgroup of $\operatorname{Perm}(S)$.

Theorem 1.1. *A set G is an abstract group iff it is a concrete group.*

Proof. $(\Longleftarrow)$: Suppose G is a concrete group. Then take composition of symmetries as the binary relation, and we see that this structure satisfies the axioms of an abstract group.

$(\Longrightarrow)$: Suppose G is an abstract group. We desire to find some object S such that G comprises the symmetries of S which preserve whatever structure is on S. Explicitly we take S to be a G-set, and consider the right action of